

NEWPORT STATE, RI, USA

WMO: 725079

Lat: 41.533N

Lon: 71.283W

Elev: 52

StdP: 100.70

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 14787

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.5	-9.9	-22.1	0.5	-11.2	-19.5	0.7	-8.3	12.1	-2.4	11.1	-3.3	6.0	300	0.497

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.6	29.2	22.7	27.7	21.9	26.5	21.3	24.3	27.3	23.5	26.0	22.8	25.0	3.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.2	25.6	22.6	17.5	24.8	22.2	16.9	24.2	73.2	27.3	70.1	26.0	67.4	25.2	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.8	7.9	DB	-15.8	32.3	3.2	1.9	-18.0	33.7	-19.9	34.8	-21.7	35.9	-24.0	37.2	(4)
(5)			WB	-16.7	25.6	3.0	1.0	-18.9	26.3	-20.6	26.9	-22.3	27.5	-24.5	28.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.7	-0.8	0.1	3.3	8.5	13.6	18.5	22.1	21.8	18.4	12.6	7.4	2.6	(6)	
	DBStd	8.87	5.41	4.60	4.23	3.42	3.35	3.18	2.52	2.48	3.10	3.94	4.30	4.73	(7)	
	HDD10.0	1248	335	278	209	67	5	0	0	0	0	20	99	234	(8)	
	HDD18.3	3091	593	511	466	294	151	36	2	3	38	181	327	489	(9)	
	CDD10.0	1518	1	1	2	23	117	256	374	366	251	102	22	4	(10)	
	CDD18.3	317	0	0	0	1	5	41	118	110	38	4	0	0	(11)	
	CDH23.3	1457	0	0	0	5	34	193	604	510	103	7	0	0	(12)	
	CDH26.7	272	0	0	0	1	6	36	133	85	10	1	0	0	(13)	
(14) Wind	WSAvg	3.6	4.2	4.2	4.4	4.0	3.5	3.1	2.9	2.8	3.1	3.6	3.9	4.0	(14)	
(15) Precipitation	PrecAvg	1176	93	88	120	105	88	91	84	96	102	99	99	111	(15)	
	PrecMax	1425	157	165	387	204	188	271	257	199	235	330	171	183	(16)	
	PrecMin	942	42	45	7	39	29	0	25	20	21	9	19	32	(17)	
	PrecStd	135	31	34	77	43	34	67	45	46	49	66	40	39	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	12.4	16.6	22.6	27.2	30.0	32.3	31.0	27.9	24.1	18.1	15.3	(19)	
		MCWB	11.3	9.7	10.6	14.3	19.3	22.3	24.5	24.0	21.9	19.0	15.6	13.5	(20)	
	2%	DB	10.7	10.1	13.0	18.1	23.4	27.4	29.7	28.8	26.0	21.3	17.0	13.5	(21)	
		MCWB	9.6	8.3	8.7	11.8	17.4	20.8	23.1	22.6	20.6	18.4	14.8	12.1	(22)	
	5%	DB	8.8	8.5	11.0	15.9	21.2	25.6	27.9	27.4	24.2	19.9	15.7	12.1	(23)	
		MCWB	7.2	6.6	8.0	10.6	16.0	20.0	22.2	22.1	20.3	17.6	13.6	11.0	(24)	
	10%	DB	7.0	7.0	9.2	13.8	19.0	23.8	26.6	26.2	23.0	18.6	14.1	10.2	(25)	
		MCWB	5.3	4.8	6.6	9.5	14.7	18.9	21.6	21.5	20.0	16.4	12.3	8.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.0	10.9	12.5	16.0	20.6	23.6	25.6	25.4	24.0	21.1	17.1	14.4	(27)	
		MCDB	12.3	11.4	14.1	19.5	25.2	28.0	30.7	29.0	26.7	22.4	17.7	14.9	(28)	
	2%	WB	9.9	8.9	10.3	13.7	18.6	22.3	24.3	24.4	22.7	19.5	15.6	12.6	(29)	
		MCDB	10.5	9.8	11.7	16.3	21.9	26.0	27.8	27.0	24.2	20.7	16.5	13.3	(30)	
	5%	WB	7.6	7.1	8.7	12.2	17.2	21.1	23.5	23.6	21.8	18.2	14.2	11.1	(31)	
		MCDB	8.6	8.1	10.2	14.2	19.9	23.9	26.3	25.8	23.2	19.3	15.2	11.7	(32)	
	10%	WB	5.4	5.0	7.3	10.8	15.8	20.2	22.7	22.9	21.0	16.9	12.8	8.8	(33)	
		MCDB	6.9	6.7	8.7	12.7	18.1	22.6	25.1	24.7	22.4	18.2	14.0	10.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.5	7.6	8.1	8.1	7.8	7.6	7.6	7.8	7.6	7.4	7.0	(35)	
		MCDBR	8.5	8.8	9.5	11.1	11.0	10.1	9.3	8.6	8.3	8.4	7.8	8.6	(36)	
	5% WB	MCWBR	8.3	7.7	6.7	6.3	6.2	5.4	4.4	4.2	5.1	6.1	7.1	8.8	(37)	
		MCWBR	8.5	8.8	8.3	8.9	9.5	8.4	7.6	6.8	6.3	7.4	7.0	8.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.304	0.312	0.335	0.372	0.413	0.457	0.482	0.440	0.373	0.343	0.314	0.302	(40)	
		taud	2.509	2.487	2.462	2.393	2.309	2.223	2.161	2.274	2.434	2.525	2.563	2.561	(41)	
	Ebn at Noon		854	902	915	899	865	824	798	823	862	854	837	828	(42)	
		Edh at Noon	73	86	99	115	128	140	147	128	102	83	69	65	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.87	2.75	3.76	4.85	5.38	5.87	6.03	5.42	4.41	2.99	2.04	1.57	(44)	
	RadStd		0.14	0.27	0.27	0.52	0.52	0.58	0.32	0.28	0.32	0.27	0.14	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	+0.79	+0.73	+0.48	N/A	N/A	+120	+76	(46)	
(47) Regional (65 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+95	+47	(47)	

Nomenclature: See separate page